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Firewalls & Network Address Translation (NAT)

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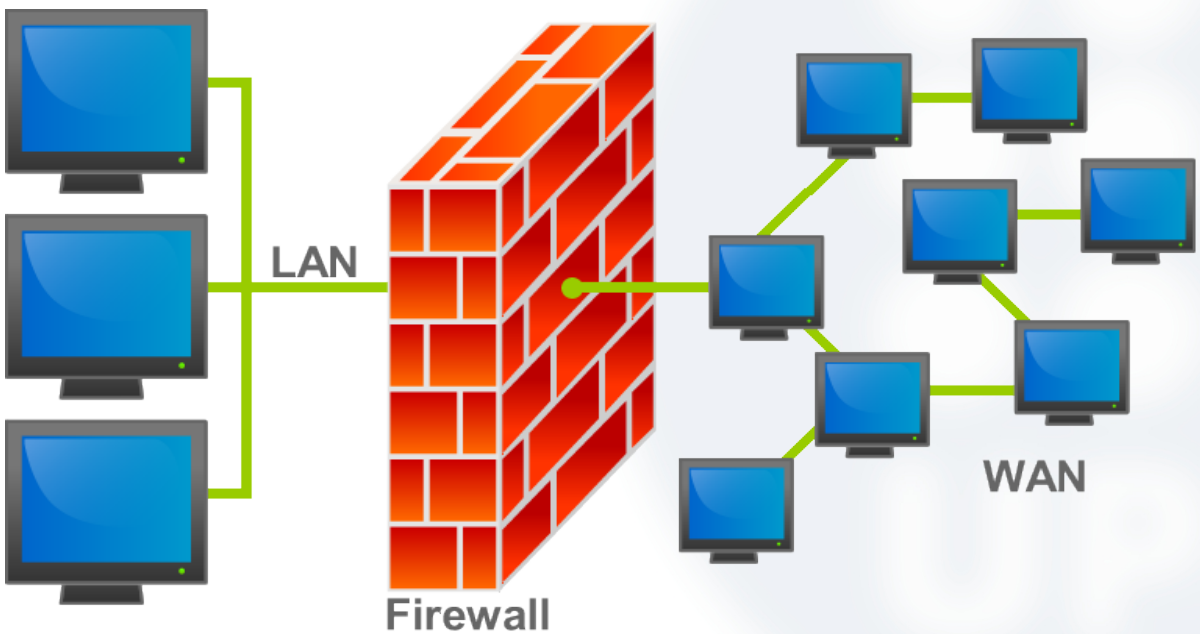
What is a firewall?

- A **firewall** is a part of a system or network that is designed to block unauthorized access while permitting authorized communications.
  - Its primary objective is to control the incoming and outgoing network traffic by ***analyzing*** the data packets and determining whether they should be allowed to pass through or not. This decision is based on a predetermined set of rules.

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What is a firewall?

- A network firewall builds a bridge between an internal network (which is assumed to be secure and trusted) and another network (usually an external network, such as the Internet, that is not assumed to be secure and trusted).



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1st generation: packet filters

- In 1988, engineers from Digital Equipment Corporation developed filter systems known as packet filter firewalls.
- Packet filters act by inspecting the packets transferred between computers:

> If a packet matches the packet filter's set of rules, the packet filter will drop (silently discard) the packet, or reject it (discard it, and send "error responses" to the source).

- This type of packet filtering pays no attention to whether a packet is part of an existing stream of traffic:
  - It filters each packet based only on information contained in the packet itself.
  - This packet filter system is "stateless".

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1st generation: packet filters (II)

Packet filter charateristics

- Works by inspecting packet headers and filtering traffic based on:
  - The IP addresses (source and/or destination).
  - The protocol (TCP, UDP, ICMP, . . .).
  - Port number (TCP, UDP) or message type (ICMP), . . .
- Low complexity.
- Fast execution.
- Available in most network devices, particularly routers.
- Does not support sophisticated rule-based models:
  - For example, they cannot detect maliciously modified parameters in an URL request.
- A stateless firewall treats each packet in isolation.
  - They have no memory of previous packets, which makes them vulnerable to some attacks.

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2nd generation: "stateful" filters

- The invention of stateful firewall is usually credited to Nir Zuk and his team at Check Point in the mid-1990s.
- Stateful firewalls are able to:
  - Track open connections.
  - Maintain a table of these open connections.
  - Associate new connection requests with existing legitimate connections.
- A stateful firewall is able to hold significant attributes of each connection in memory, including:
  - IP addresses.
  - Ports.
  - Sequence numbers.

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3rd generation: Application Layer

- A network-based application layer firewall is a firewall operating at the application layer of a protocol stack
- Application layer filtering can “understand” certain application protocols. It is able to detect:
  - If an unwanted protocol is attempting to bypass the firewall on an allowed port.
  - If a protocol is being abused in any harmful way.

http firewall

For example, an application layer firewall with http support is able to block the downloading of a specific web page in a site, while allowing the downloading of other pages in the same site.

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Reasons to use NAT

- Problem:
  - Public IP addressing is limited (now exhausted!) and expensive.
- Solution:
  - Using private IP adressing (RFC 1918) in internal networks.
  - Available adress blocks:
    - A class: 10.0.0.0/8.
    - B class range: 172.16.0.0/16-172.31.0.0/16.
    - C class range: 192.168.0.0/16.
  - Problem: private IP addresses cannot be used in public networks (like the Internet).

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NAT introduction

Definition

NAT: Network Address Translation

- NAT is a mechanism that translates IP addresses (and port numbers).
- NAT allows to connect public and private networks, and to save public IP addressing.
- Supports TCP, UDP and ICMP.
- The specifications of NAT can be found in RFC 1631.
- When using NAT, two address translations are performed: one translation when the packet departs the *NAT router*[1], and the reverse translation when the packet returns to the NAT router.

[1] For simplicity, sometimes we will use the term *router* instead of *NAT route*

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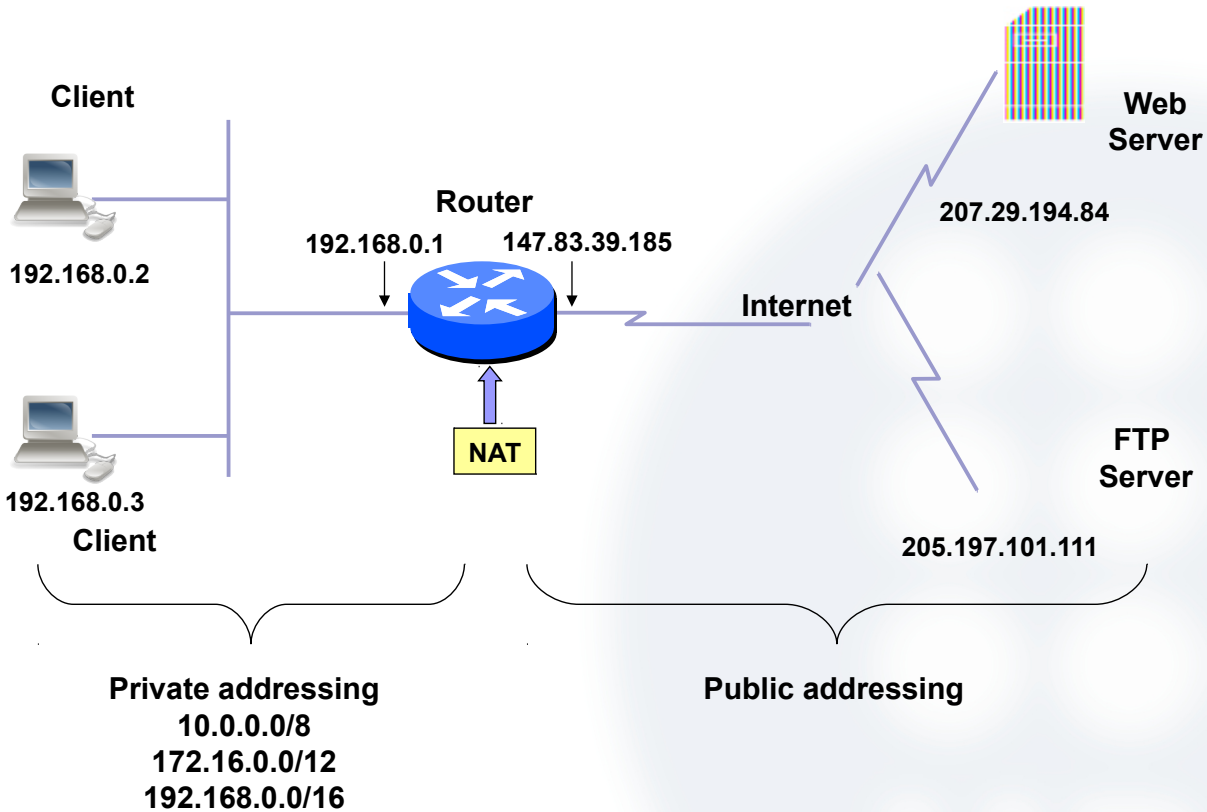
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NAT usage



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NAT types

- Two main types:
  - Source NAT (SNAT)
    - The router translates the source address.
    - This translation is performed after routing, just before the packet leaves the router.
    - The router will perform the reverse operation (translation of destinarion address) when response packets arrive.
  - Destination NAT (DNAT)
    - The router translates the destination address.
    - This translation is performed before routing, just when the packet arrives to the router.
    - The router will perform the reverse operation (translation of source address) when response packets arrive.

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Consequences of NAT usage

- When IP address is changed:
  - Whenever a source or destination address is changed, the checksums must be recalculated.
  - This includes IP, TCP and UDP checksums[2].
- When NAT is used, response packets should return to the same NAT router (to perform the reverse translation).
- For this reason, in most of the cases there is only one NAT router to manage the Internet access (no matter if this router have connections with multiple ISPs).

[2]Notice that TCP and UPD checksums are calculated including the IP pseudo-header.

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Problems and limitations I

- When using NAT, the Internet (tipically a connectionless network) becomes a pseudo-connection-oriented network.
- NAT breaks with the transparency of protocol layers.
- The NAT router becomes a bottleneck.
  - It must manage the translations of all ingoing and outgoing connections: problems in terms of CPU and memory.
  - The number of available public addresses is limited, and this also limits the number of possible external connections.

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Problems and limitations II

- Some protocols and applications include IP addresses (and port numbers) in their headers and data fields.
- These IP addresses must also be translated for a correct behavior. The router will need to use not only NAT but also ALG (Application-Level Gateway).
- Examples:
  - ICMP Destination unreachable messages include an IP address in the Data field.
  - Some application protocols (H.323, NetBIOS) have IP addresses in different parts of the Data field.

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FTP+NAT example I

- At the beginning of a FTP session there is an exchange of some control commands, and the IP addresses of the host are included in these commands.
- These addresses are included in the Data field, and NAT must translate them to make FTP properly work.
- Addresses are written in ASCII text, so addresses are not codified in 32 bits.
- The objective is to maintain the lenght of the original addresses, so the router does not need to change the sequence numbers of the connection.

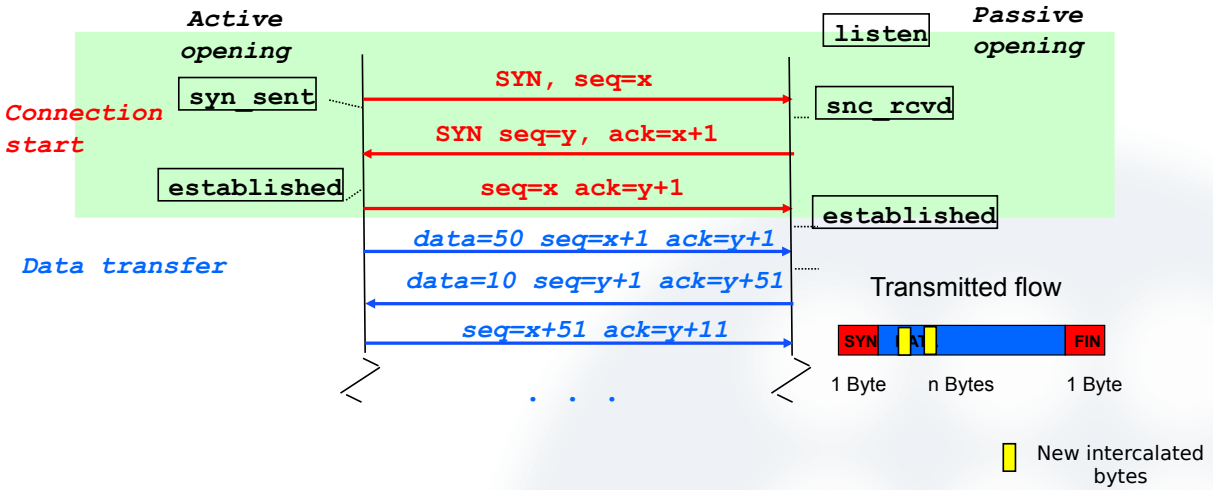
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FTP+NAT example II

- If the new address is less long that the old one (in ASCII), the new address is filled with 0's.
  - Example: if NAT translates 206.245.160.2 to 192.168.1.2, the router fixes the problem by adding 0's until having the same number of characters. So, the router will codify the address as 192.168.001.2.
- If the new address is longer than the old one, some bytes must be added.
  - Example: if NAT translates 192.168.1.2 to 206.245.160.2, the router fixes the problem by adding 2 extra bytes.

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FTP+NAT example III



- The router must change consequently all sequence and ACK numbers to maintain the connection coherence.
- This increases complexity, because the NAT router must manage some additional state information for every FTP connection.

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Conclusions

- NAT has extended the lifetime of IPv4, delaying IPv6 deployment.
- NAT is far from being an optimal solution.
- There can be different behaviors for different NAT implementations.
- New protocols and applications will need extra NAT support.
- Use NAT only if it is necessary.

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Netfilter I

- Netfilter is the packet filtering framework used in Linux platforms.

> Packet processing occurs **INSIDE** the Linux kernel

- The architecture of Netfilter contains *chains of rules* for the treatment of packets.
- Each *chain* is associated with a different “role“ of the host when processing packets.
- Packets are processed by sequentially traversing the rules into the chains.
- Every network packet arriving at or leaving from the computer traverses at least one chain.

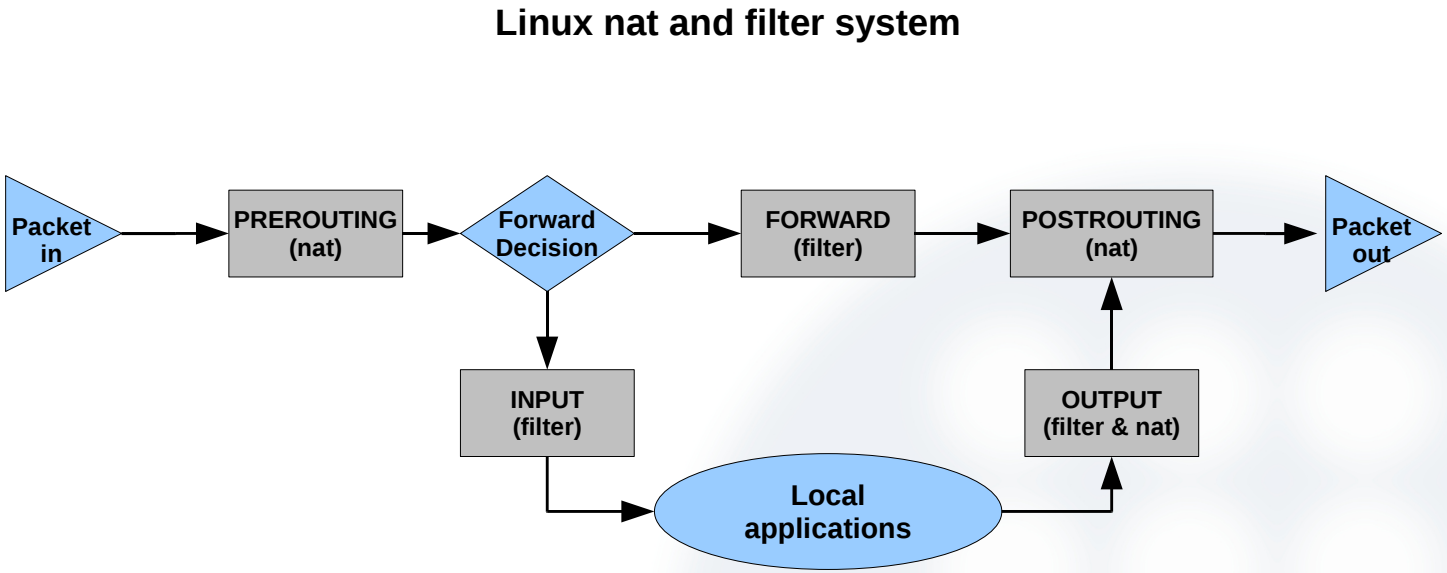
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Netfilter II

- There are five predefined chains:
  - **PREROUTING**: this is the first chain that a packet coming from the network finds. This chain is located before the routing decision.
  - **INPUT**: this chain applies when the packet is going to be locally delivered (the host is the destination).
  - **FORWARD**: the packet will enter this chain if it comes from the network, but the host is not the destination (the host acts as router).
  - **OUTPUT**: this chain applies when packets are sent by the host (the host is the origin).
  - **POSTROUTING**: the packet will enter this chain after the routing decision has been made.

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Netfilter III



- Predefined chains have a default policy, (for example **DROP**), which is applied to the packet if it reaches the end of the chain (not matching any rule).
- A chain may be empty of any rule.

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Rules in Chains I

- When a packet enters a chain, the kernel verifies if the rules within the chain match this packet.
- Each rule contains some specifications to decide if the packets match it.
- Each rule also contains a verdict to be applied in case of matching the specifications.
- Possible verdicts depend on the chain: accept or drop for filtering, change addresses (and ports) for NAT.
- Operation:
  1. Rules are examined sequentially.
  2. If a rule DOES NOT match the packet, the next rule is examined.
  3. If a rule DOES match the packet, the action indicated in the verdict is taken, and no more rules are examined.

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Rules in Chains II

4. If a packet reaches the end of the chain without matching any rule, the default chain policy is applied to the packet.

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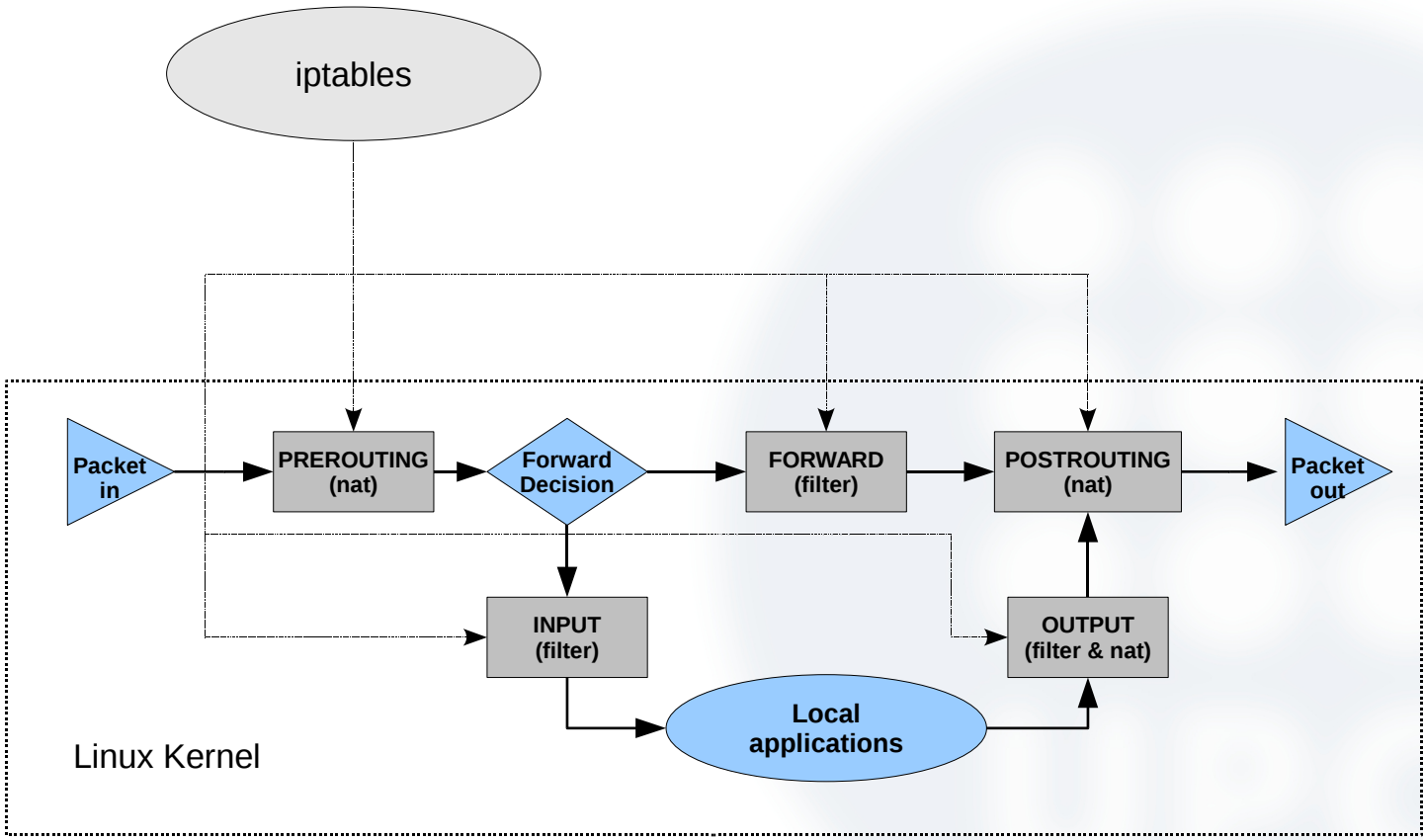
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iptables I

How to configure the netfilter chains in the Linux Kernel?
Using the `iptables` command.



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iptables II

syntax

```
iptables <table> <op> <chain> <pkt-match-condition> <action>
```

- **table**: Selects the table to work with
  - `-t filter`: selects packet filtering (or nothing, this is the default option).
  - `-t nat`: selects NAT table.
- **op**: Rule operations within a chain:
  - APPEND a new rule to the chain (-A).
  - INSERT a new rule in a position within a chain (-I).
  - DELETE a rule in a position within a chain (-D).
  - MOVE a rule from a position to another position (-R).
- **chain**: The chain name to operate with
  - Packet filtering: INPUT, OUTPUT and FORWARD.
  - NAT: PREROUTING, POSTROUTING and OUTPUT.

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iptables III

- **Packet Matching Condition**: The set of conditions that a packet must satisfy. These conditions can be:
  - Physical/link layer conditions: network interface where the packet is received/transmited.
  - Network layer conditions: fields in the IP header.
  - Transport layer conditions: fields in the transport header.
- **Action**: the verdict is applied to the packet if all the rule conditions are satisfied.

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Conditions for physical/link layer

The only conditions on these layers are the network interface where the packet has to be received or has to be transmited.
Syntax:

- `-i, --in-interface [!] name`
  Name of an interface via which a packet was received (only for packets entering the INPUT, FORWARD and PREROUTING chains). When the "!" argument is used before the interface name, the sense is inverted.
- `-o, --out-interface [!] name`
  Name of an interface via which a packet is going to be sent (for packets entering the FORWARD, OUTPUT and POSTROUTING chains).

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Conditions for the network layer

The conditions of the network layer are based on some fields of the IP header:

- `-p, --protocol [!] protocol`
  The protocol of the packet to check. The specified protocol can be one of `tcp`, `udp`, `icmp`, or `all`, or it can be a numeric value, representing one of these protocols or a different one.
- `-s, --source [!] address[/mask]`
  Source specification. Address can be either a network name, a hostname, a network IP address (with /mask), or a plain IP address. A “!” argument before the address specification inverts the sense of the address.
- `-d, --destination [!] address[/mask]`
  Destination specification. Address can be either a network name, a hostname, a network IP address (with /mask), or a plain IP address.

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Conditions for ICMP

If `--protocol icmp` is specified in the conditions of the network layer:

- `--icmp-type [!] typename` This allows specification of the ICMP type, which can be a numeric ICMP type, or one of the ICMP type names shown by the command
  `iptables -p icmp -h`

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Conditions for the transport layer: TCP I

If `--protocol tcp` is specified in the conditions of the transport layer:

- `--source-port [!] port[:port]`
  Source port or port range specification.
- `--destination-port [!] port[:port]`
  Destination port or port range specification.
- `--tcp-flags [!] mask comp`
  Match when the TCP flags are as specified. The first argument is the flags which we should examine, written as a comma-separated list, and the second argument is a comma-separated list of flags which must be set. Flags are: SYN ACK FIN RST URG PSH ALL NONE.

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Conditions for the transport layer: TCP II

- `[!] --syn`
  Only match TCP packets with the SYN bit set and the ACK,RST and FIN bits cleared. Such packets are used to request TCP connection initiation; for example, blocking such packets coming in an interface will prevent incoming TCP connections, but outgoing TCP connections will be unaffected. It is equivalent to --tcp-flags SYN,RST,ACK,FIN SYN. If the “!” flag precedes the “--syn”, the sense of the option is inverted.
- `--tcp-option [!] number`
  Match if TCP option set.

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Conditions for Transport Layer: UDP

If `--protocol udp` is specified in the conditions of the transport layer:

- `--source-port [!] port[:port]` Source port or port range specification.
- `--destination-port [!] port[:port]` Destination port or port range specification.

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Verdict and Actions I

When a packet matches the conditions of a rule, the verdict is positive and an acction has to be applied to the packet. The action is specified, in the command line, with a "`-j`":

syntax

```
iptables <table> <op> <chain> <pkt-condition> -j action
```

- For filter table (**-t filter**), actions are:
  - **DROP**: the packet is silently discarded and it disapears from the system.
  - **ACCEPT**: the packet is accepted by the rule and continues to the next chain.

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Verdict and Actions II

- For nat table (**-t nat**), actions have an argument:
  - **DNAT**: it is only valid in the PREROUTING and OUTPUT chains. It specifies that the destination address of the packet should be modified.
    One option:
    `--to-destination`
    `[ipaddr][-ipaddr][:port-port]`
    which can specify a single new destination IP address, an inclusive range of IP addresses, and optionally, a port range (which is only valid if the rule also specifies -p tcp or -p udp). If no port range is specified, then the destination port will never be modified. If no IP address is specified then only the destination port will be modified.

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Verdict and Actions III

- **SNAT**: it is only valid in the nat table, in the POSTROUTING chain. It specifies that the source address of the packet should be modified.
  One option:
  `--to-source ipaddr[-ipaddr][:port-port]`
  which can specify a single new source IP address, an inclusive range of IP addresses, and optionally, a port range (for -p tcp or -p udp). If no port range is specified then where possible, no port alteration will

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iptables examples I

- Adding a rule in the INPUT chain to drop packets whose source IP address is 192.168.1.1 and transport protocol TCP:

```
iptables -t filter -A INPUT -s
192.168.1.1 -p tcp -j DROP
```

- Adding a rule in FORWARD to drop packets whose source IP addresses are in range 192.168.1.0/24, the protocol ICMP and icmp type echo-request, and received in the eth0 network interface:

```
iptables -t filter -A FORWARD -i eth0 -s
192.168.1.0/24 -p ICMP --icmp-type
echo-request -j DROP
```

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iptables examples II

- Adding a rule in OUTPUT to avoid outgoing http connections towards the www.google.com server:

```
iptables -t filter -A OUTPUT -d
www.google.com -p tcp --dport 80 --syn
-j DROP
```

- Adding a rule to do SNAT for all packets that leave the router using the eth1 interface, translating to the IP address 172.16.1.1:

```
iptables -t nat -A POSTROUTING -o eth1
-j SNAT --to 172.16.1.1
```